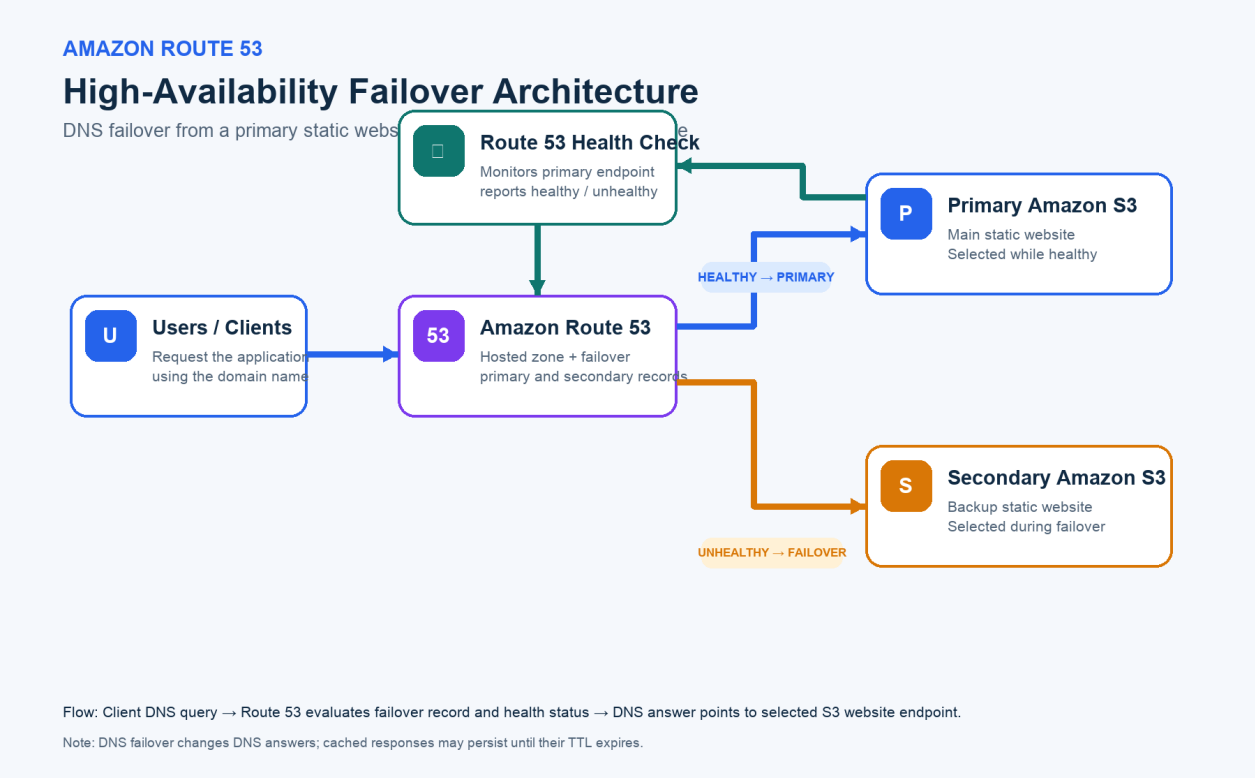
AMAZON ROUTE 53

Failover Routing for High Availability

Combined project package

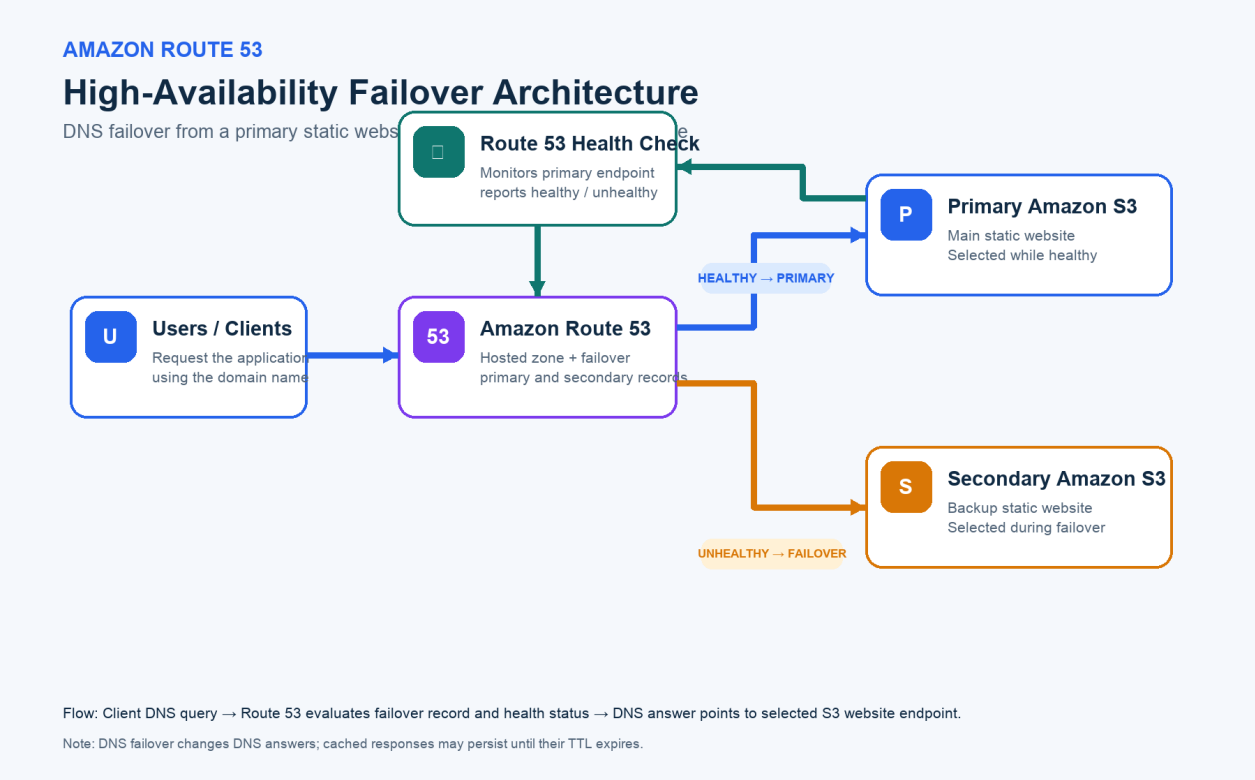
Primary website • Secondary failover website • Architecture • Implementation report



Project code: 24CC3014-P044 **Team ID:** Y24-SAA-Team185

Architecture and traffic flow

The Route 53 health check observes the primary website endpoint. Route 53 uses the health status with its failover records to return the primary endpoint during normal operation and the secondary endpoint when the primary is unhealthy. The diagram separates health monitoring from DNS routing decisions.



DNS behavior: Route 53 changes DNS responses based on health and failover configuration. Recursive resolvers and clients may continue using cached answers until their TTL expires.

Website demonstrations included

Both source folders contain a static HTML/CSS/JavaScript demonstration of the same failover architecture. The pages show the project state at load time; the primary site also exposes a frontend simulation control.

Source folder	Demonstration state	Included website files
AWS-Project-Primary	Primary healthy; normal traffic targets Primary S3. Secondary is standby.	index.html, style.css, script.js
AWS-Project-Secondary	Primary unhealthy; failover traffic targets Secondary S3.	index.html, style.css, script.js

End-to-end behavior

Normal operation: Client resolves the domain → Route 53 sees the primary as healthy → DNS answer targets the primary S3 website endpoint.

Failover: Health check marks the primary endpoint unhealthy → Route 53 selects the secondary failover record → DNS answer targets the secondary S3 website endpoint.

Recovery: After the primary endpoint is healthy again, Route 53 can resume primary answers according to the configured failover records and DNS caching behavior.

The supplied nine-page implementation report follows this project-specific overview. It is preserved as provided.

AWS PROJECT

AMAZON ROUTE 53 FAILOVER ROUTING FOR HIGH AVAILABILITY

AWS Deployment & Implementation Report

Project Code: 24CC3014-P044

Team ID: Y24-SAA-Team185

Team Members

	Name	Section
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Deployment scope

A high-availability static web application using two Amazon S3 website endpoints, Amazon Route 53 failover routing, and a Route 53 health check. When the primary endpoint becomes unhealthy, DNS traffic can be directed to the secondary endpoint.

Technology: Amazon Web Services (AWS)

Prepared for project demonstration • 23 September 2026

1. Executive Summary

This project demonstrates an automatic failover mechanism for a web application using Amazon Route 53. Two static websites are hosted using Amazon S3: one is the primary website and the other is the secondary backup website. Route 53 provides DNS-based failover routing, while a Route 53 health check monitors the primary endpoint.

Under normal conditions, users are directed to the primary website. If the primary endpoint becomes unhealthy, Route 53 can route DNS requests to the secondary website. This reduces dependence on a single endpoint and helps maintain application availability.

> **Target outcome**
>
> Primary endpoint healthy → traffic goes to Primary S3 website. Primary endpoint unhealthy → Route 53 failover routes traffic to Secondary S3 website.

High-level architecture

	Role
Users / Clients	Access the application using the configured domain name.
Amazon Route 53	Manages DNS and failover routing.
Route 53 Health Check	Checks the availability of the primary endpoint.
Primary Amazon S3	Hosts the main static website.
Secondary Amazon S3	Hosts the backup static website.
CloudWatch / SNS	Optional monitoring and failure notifications.

2. Objectives and Scope

- Deploy a primary web endpoint for the application.
- Create a secondary endpoint that is ready to act as a backup.
- Configure Route 53 failover routing with primary and secondary records.
- Configure a health check to monitor the primary endpoint.
- Automatically direct DNS traffic to the secondary endpoint when the primary is unhealthy.
- Test the failover process by simulating primary-endpoint failure.
- Improve availability and reduce service downtime.

3. Problem Statement

A web application may become unavailable when its primary server or endpoint has a server, network, or application problem. Without an automated backup mechanism, users may be unable to access the application until the issue is fixed or a person manually switches to another endpoint.

Key problem

When the primary endpoint fails, there may be no automatic way to send users to an available backup endpoint.

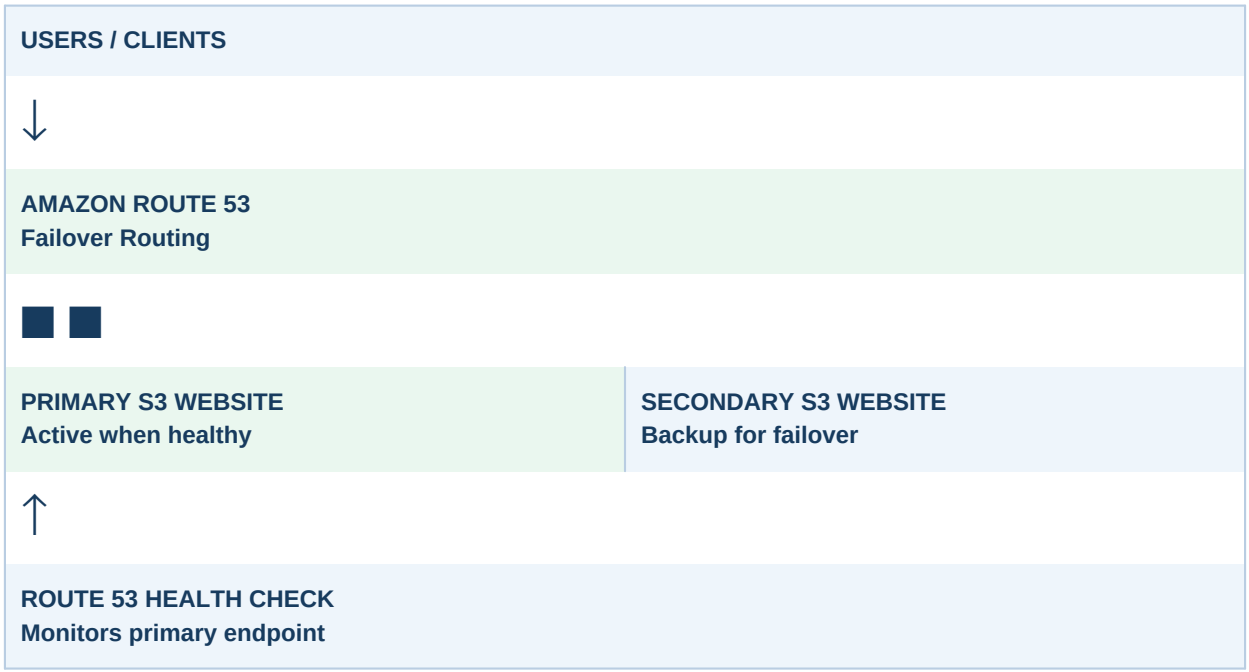
Problem flow

USER	→	PRIMARY WEBSITE	→	FAILURE
Requests application		Main S3 endpoint		Primary becomes unavailable

Proposed solution

Route 53 is configured with a primary failover record and a secondary failover record. The health check monitors the primary endpoint. If it is healthy, users are routed to the primary. If it is unhealthy, Route 53 uses the secondary record for failover.

4. Proposed Architecture



5. AWS Services Used

	Purpose in the project
Amazon Route 53	DNS management and failover routing between primary and secondary endpoints.
Route 53 Health Check	Monitors the availability of the primary endpoint.
Amazon S3	Hosts the primary and secondary static websites.
Amazon CloudWatch	Optional monitoring and metrics for AWS resources.
Amazon SNS	Optional notifications about failures or important events.

Core services

> **Route 53 + Health Check + S3**
>
> These are the main components required for the failover demonstration.

6. Prerequisites

- An AWS account with access to Route 53 and S3.
- A domain name or hosted zone suitable for the DNS configuration.
- Two S3 buckets configured to host the primary and secondary static websites.
- A primary endpoint that can be checked by Route 53 Health Check.
- Website files for both the primary and backup pages.
- A test plan for simulating primary-endpoint failure.

7. Create the Primary Website

- Create an S3 bucket for the primary website.
- Enable static website hosting as required by the chosen S3 website configuration.
- Upload the main website files.
- Record the primary website endpoint for Route 53 configuration.

8. Create the Secondary Website

- Create a second S3 bucket for the backup website.
- Upload the same or equivalent website files.
- Keep the secondary endpoint available for failover.
- Record the secondary website endpoint.

9. Configure Amazon Route 53

Create or use the hosted zone for the project domain and configure two failover records. The primary record points to the primary endpoint and the secondary record points to the backup endpoint.

	Role	Target
Primary failover record	Normal traffic	Primary S3 website endpoint
Secondary failover record	Failover traffic	Secondary S3 website endpoint

10. Configure Route 53 Health Check

- Create a health check for the primary endpoint.
- Use the health check to determine whether the primary endpoint is available.
- Associate the health check with the primary failover record.
- Route 53 uses the health status when deciding whether normal primary routing can continue.

> **Important distinction**
>
> The Health Check detects the primary endpoint's health. Route 53 is the component that performs the DNS failover decision.

11. How the System Works

1	User enters the application domain name.
2	DNS request reaches Amazon Route 53.
3	Route 53 checks the primary record and its health status.
4	If the primary is healthy, traffic is directed to the primary S3 website.
5	If the primary is unhealthy, Route 53 directs traffic to the secondary S3 website.
6	When the primary recovers, normal primary routing can resume according to the configured failover and DNS behavior.

12. Implementation Procedure

	Implementation activity
1	Create the primary S3 static website and verify it works.
2	Create the secondary S3 static website and verify it works.
3	Configure the Route 53 hosted zone/domain records.
4	Create the primary and secondary failover records.
5	Create and configure the Route 53 Health Check for the primary endpoint.
6	Test normal access through the primary endpoint.
7	Simulate a primary-endpoint failure.
8	Verify that DNS failover uses the secondary endpoint.

13. Testing and Verification

	Condition	Expected result
Primary test	Primary endpoint is healthy	Primary website is accessible.
Failure detection	Primary endpoint becomes unavailable	Health Check detects unhealthy status.
Failover test	Primary is unhealthy	Route 53 uses the secondary failover record.
Recovery test	Primary endpoint is restored	Primary endpoint can be used again according to routing/health status.

Metrics to observe

- Failure detection time.
- Failover behavior and DNS response.
- DNS TTL effects during testing.
- Application availability before and after failure.

Expected result

When the primary endpoint becomes unavailable, Route 53 should detect the unhealthy state and direct DNS traffic toward the secondary endpoint, reducing application downtime.

14. Security Implementation

	Recommended practice
IAM	Use appropriate permissions for the AWS resources involved.
S3 security	Configure bucket permissions carefully and avoid unnecessary public access.
HTTPS / SSL	Use secure website communication where applicable to the deployment design.
AWS credentials	Never expose or share AWS access keys.
Least privilege	Give users and AWS services only the permissions they need.
Monitoring	Monitor AWS resources for unusual activity or configuration issues.

15. Problems and Solutions

	Solution
Primary endpoint unavailable	Use Route 53 Failover Routing.
Need to detect primary failure	Configure a Route 53 Health Check.
Manual switching is slow	Automate failover through Route 53.
Backup website not ready	Maintain a tested secondary S3 website.
Need monitoring	Use CloudWatch where required.
Need failure notifications	Use SNS where required.

16. Benefits

- Automatic failover between primary and secondary endpoints.
- Reduced dependence on a single website endpoint.
- Health monitoring of the primary endpoint.
- No manual DNS switching during the failover event.
- Simple architecture suitable for a student AWS demonstration.
- Improved application availability.

17. Troubleshooting Checklist

	Area to inspect
Primary website works but failover does not occur	Failover record type, health-check association, record targets, and DNS configuration.
Health Check reports unhealthy unexpectedly	Primary endpoint availability and health-check settings.
Secondary website cannot be accessed	Secondary S3 website configuration and endpoint.
DNS still reaches primary during testing	DNS TTL/cache and current Route 53 health/routing status.
Domain does not resolve	Hosted zone, name-server delegation, and DNS record configuration.

18. Demonstration Procedure

- Open the application domain and confirm the primary website is displayed.
- Confirm the primary endpoint is healthy.
- Temporarily make the primary endpoint unavailable in a controlled test.
- Wait for the health status and DNS behavior to reflect the failure.
- Access the application again and verify the secondary website is used.
- Restore the primary endpoint and verify the health status returns to normal.

Demonstration result

The successful demonstration should show the complete flow: normal primary access → simulated primary failure → health detection → Route 53 failover → secondary website access.

19. Project Outcome

The project provides an automated DNS failover mechanism for a web application. The primary and secondary S3 endpoints provide two destinations, while Route 53 and its health check work together to detect primary-endpoint availability and route traffic accordingly.

20. End-to-End Project Checklist

■ Primary S3 website created.	■ Secondary S3 website created.
■ Both website endpoints tested.	■ Route 53 hosted zone/domain configuration completed.
■ Primary failover record configured.	■ Secondary failover record configured.
■ Route 53 Health Check created for the primary.	■ Normal primary routing verified.
■ Primary failure simulated safely.	■ Secondary failover verified.
■ Primary recovery verified.	■ Optional CloudWatch/SNS monitoring considered.

21. Conclusion

Amazon Route 53 provides DNS-based failover routing for the project. A primary and secondary S3 website are configured as the two endpoints. Route 53 Health Check monitors the primary endpoint, and when the primary becomes unhealthy, Route 53 can fail over DNS traffic to the secondary endpoint. This helps reduce downtime and improves availability.

Final outcome

An automated failover mechanism is demonstrated using Amazon Route 53 + Health Check + two Amazon S3 website endpoints.

22. Team Details

	Name	Section
2400033275	PADALA GOWTHAM AVINASH REDDY	S-51-MA
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23. References and Source Notes

• AWS Route 53: https://aws.amazon.com/route53/

• AWS Route 53 Developer Guide: https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/

• Amazon S3: https://aws.amazon.com/s3/

• AWS Route 53 Health Checks: https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/health-checks-creating.html

Project Code: 24CC3014-P044 | Team ID: Y24-SAA-Team185